DukaanSetu – Multi-Vendor E-commerce Platform

Presented By :- Harsh Patel

Project Overview

DukaanSetu is acomprehensive multi-vendore-commerce platform builtusing the MERN stack,designed to bridge the digitaldivide for local shopkeepers.

The application connectstraditional brick-and-mortarbusinesses with onlinecustomers, creating newrevenue streams whileproviding users withconvenient access to avariety of local products.

The platform features threedistinct user panels—Admin,Shopkeeper, and Customer—each with tailoredfunctionalities to efficientlymanage products, orders,and overall platformoperations.

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System Scope & Capabilities

Multi-Vendor Platform

Provides online presence for local shops with product listings and inventory management

Role-Based Access

Supports multiple user types with distinct permissions and dashboard views

Smart Search & Filter

Enables customers to browse and find products efficiently across vendors

Unified Cart System

Multi-vendor cart allows adding products from different shopkeepers in one checkout

Secure Authentication

JWT-based security ensures protected user accounts and data privacy

Order Tracking

Complete order lifecycle management with review and feedback system

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System Architecture & Modules

The platform comprises five core modules working in harmony to deliver seamless e-commerce functionality:

01

Authentication Module

User registration and login with JWT-based security and role-based access control for secure platform access

02

Customer Module

Product search and filtering, multi-vendor cart management, order history tracking, and review submission capabilities

03

Shopkeeper Module

Product catalog management including adding, updating, and deleting items, along with order tracking and earnings analytics

04

Admin Module

Comprehensive platform oversight with user and category management, activity monitoring, and content moderation tools

05

Backend Module

RESTful APIs for data exchange, MongoDB database management, and server-side business logic implementation

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Technology Stack

Frontend

React.js

Component-based UI framework for dynamic, responsive user interface

Bootstrap 5

CSS framework ensuring consistent, mobile-first design across devices

CSS3

Styling language for custom animations and visual enhancements

Backend

Node.js

Server runtime enabling JavaScript-based backend development

Express.js

Web framework for building RESTful API endpoints efficiently

Bcrypt.js

Password hashing library ensuring secure credential storage

Database

MongoDB

NoSQL database providing flexible schema design for storing user data, product catalogs, orders, and platform analytics. Offers scalability and efficient document-based data management.

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Key Advantages

Empowers Local Businesses

Helps small shopkeepers establish digital presence and reach wider customer base without technical expertise

User-Friendly Interface

Intuitive design ensures easy navigation for all user types, reducing learning curve and improving adoption

Secure Authentication

JWT-based security with encrypted passwords protects sensitive user information and transaction data

Unified Shopping Experience

Multi-vendor cart system allows customers to purchase from multiple shopkeepers in single transaction

Time Efficiency

Reduces shopping time for customers whilst providing shopkeepers with automated inventory and order management

Centralized Management

Admin panel provides complete oversight of platform activity, users, and content moderation

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Business Impact & Benefits

Economic Growth

Supports local economy by enabling small businesses to expand digital footprint and increase revenue potential

Sales Growth

Increases sales opportunities for vendors by providing 24/7 online storefront accessible to wider audience

Digital Presence

Transforms offline shops into digital businesses, ensuring competitiveness in modern marketplace

Customer Convenience

Enhances shopping experience with easy product discovery, secure transactions, and home delivery options

Trust Building

Review and rating system establishes credibility and helps customers make informed purchasing decisions

Platform Independence

Reduces dependency on large e-commerce platforms, giving shopkeepers control over their business

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Future Enhancements

The platform can be extended with advanced features to further enhance functionality and user experience:

Payment Integration
Implement Razorpay or Stripe for secure online transactions, enabling seamless payment processing

Cloud Storage
Integrate AWS S3 or Cloudinary for scalable image hosting and faster content delivery

Multi-Language Support
Add language translation to reach diverse customer base and expand market reach

Real-Time Notifications
Implement email and SMS alerts for order updates, stock changes, and promotional messages

Chat System
Enable direct communication between customers and shopkeepers for queries and support

Advanced Analytics
Provide shopkeepers with detailed sales reports, customer insights, and performance metrics

Stock Alerts
Automated low-stock notifications help shopkeepers manage inventory levels proactively

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Project Highlights

3

User Panels

Admin, Shopkeeper, and Customer dashboards

5

Core Modules

Complete system functionality

100%

MERN Stack

Full JavaScript ecosystem

Project Goal

Built to connect local shopkeepers with online customers through a multi-vendor platform, enabling small businesses to establish digital presence and reach wider audiences

Technical Approach

Developed using MongoDB for flexible data storage, Express.js and Node.js for RESTful APIs, and React.js with Bootstrap 5 for responsive user interfaces

Key Features

JWT-based authentication, multi-vendor shopping cart, role-based access control, product catalog management, order tracking system, and review functionality

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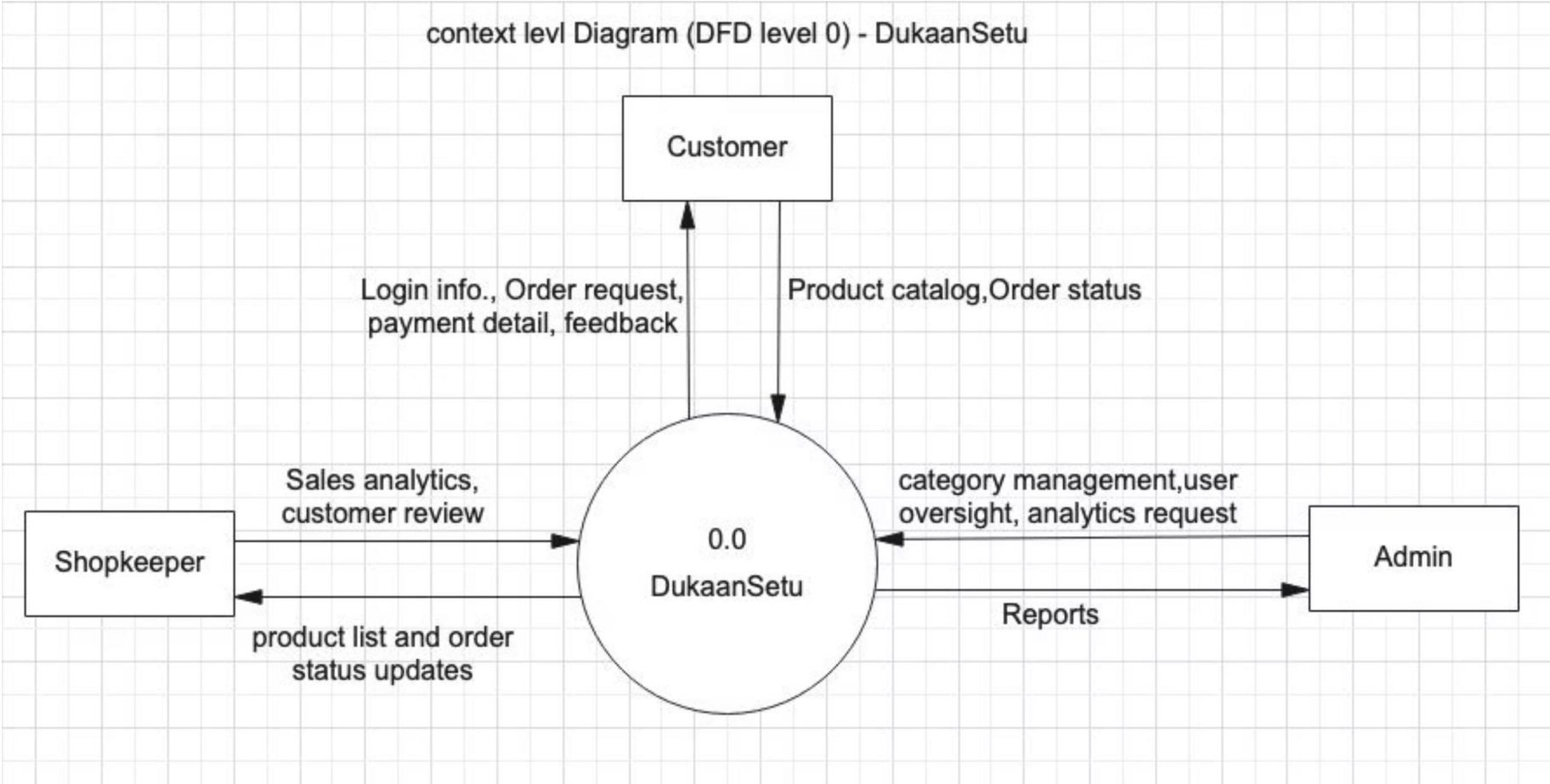
Context level Diagram



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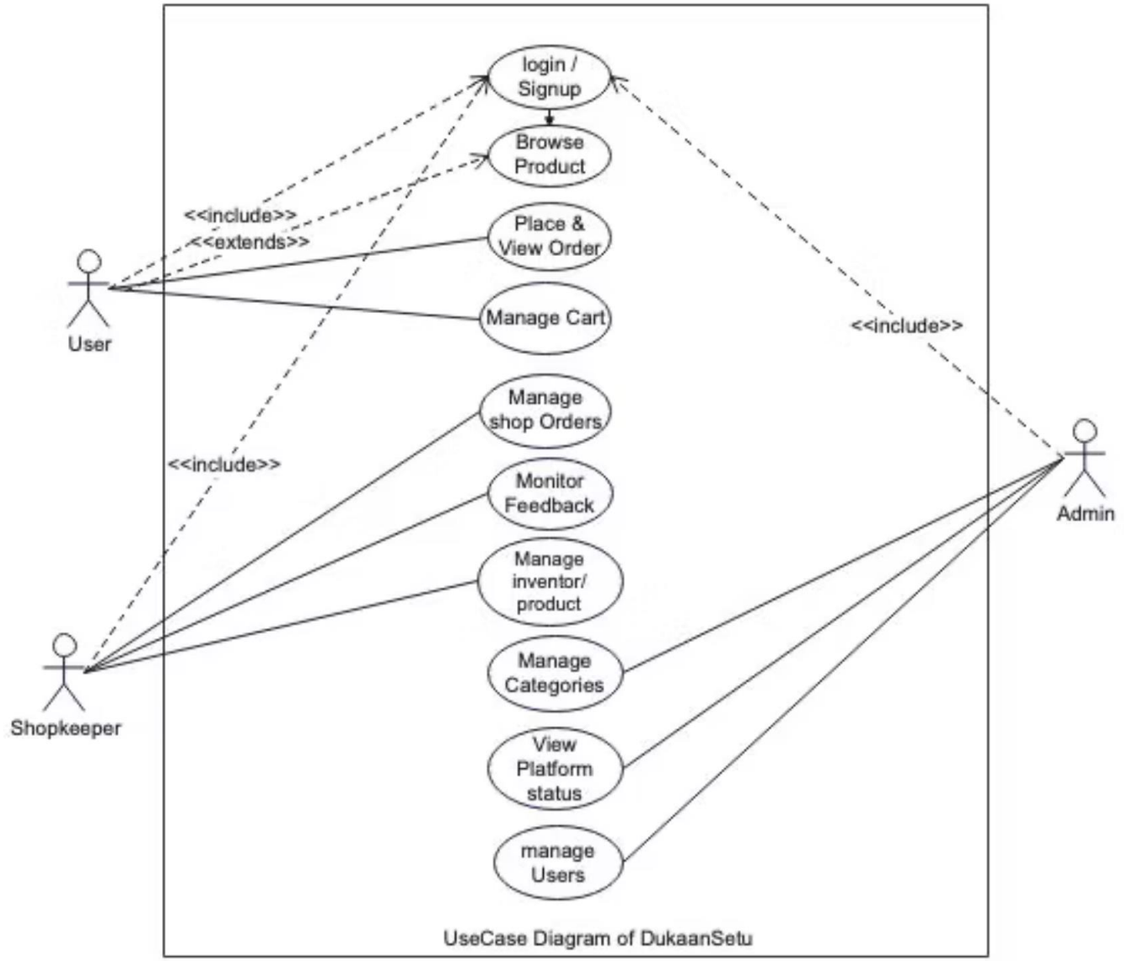
UseCase Diagram



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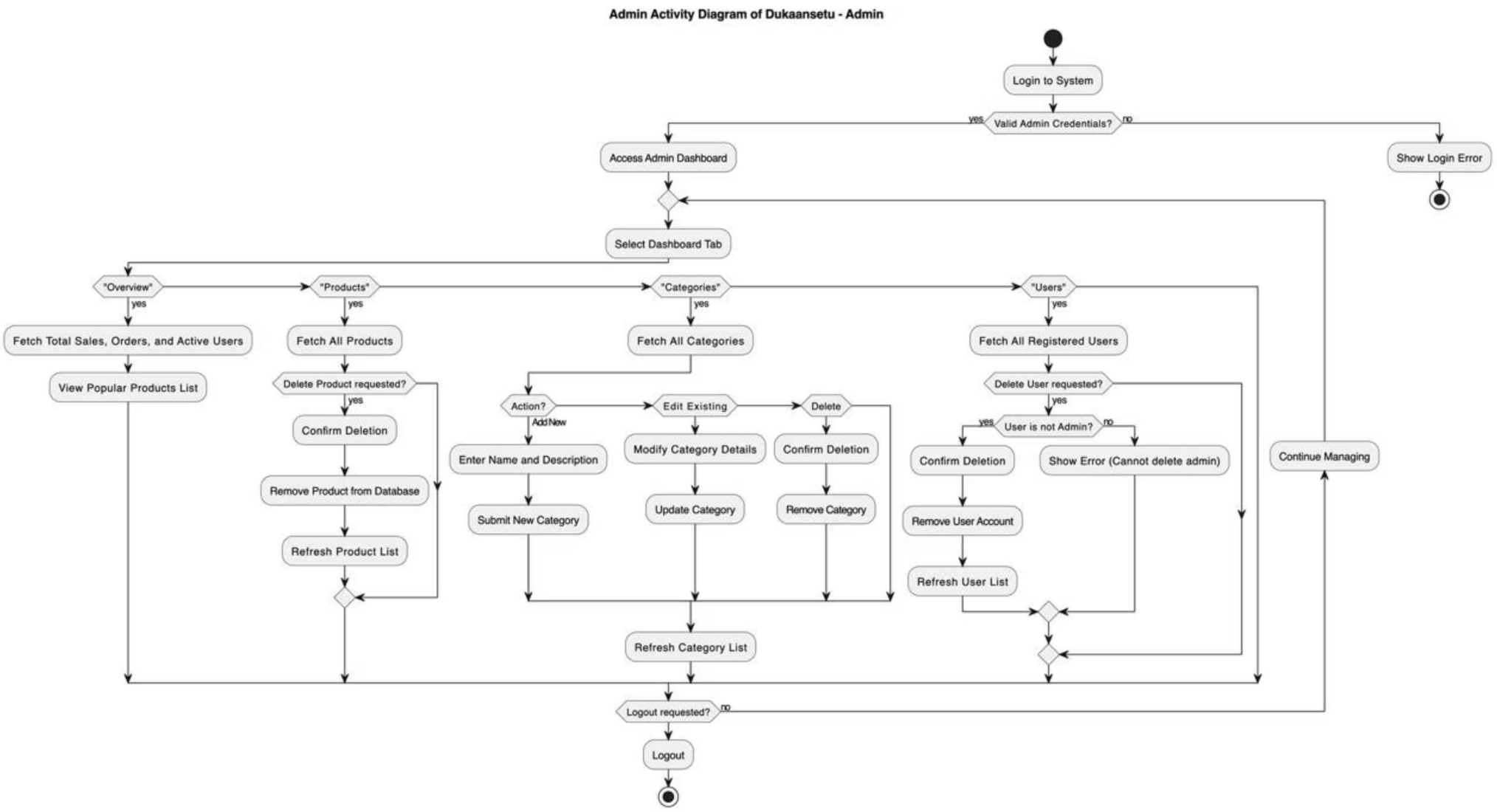
Activity Diagram



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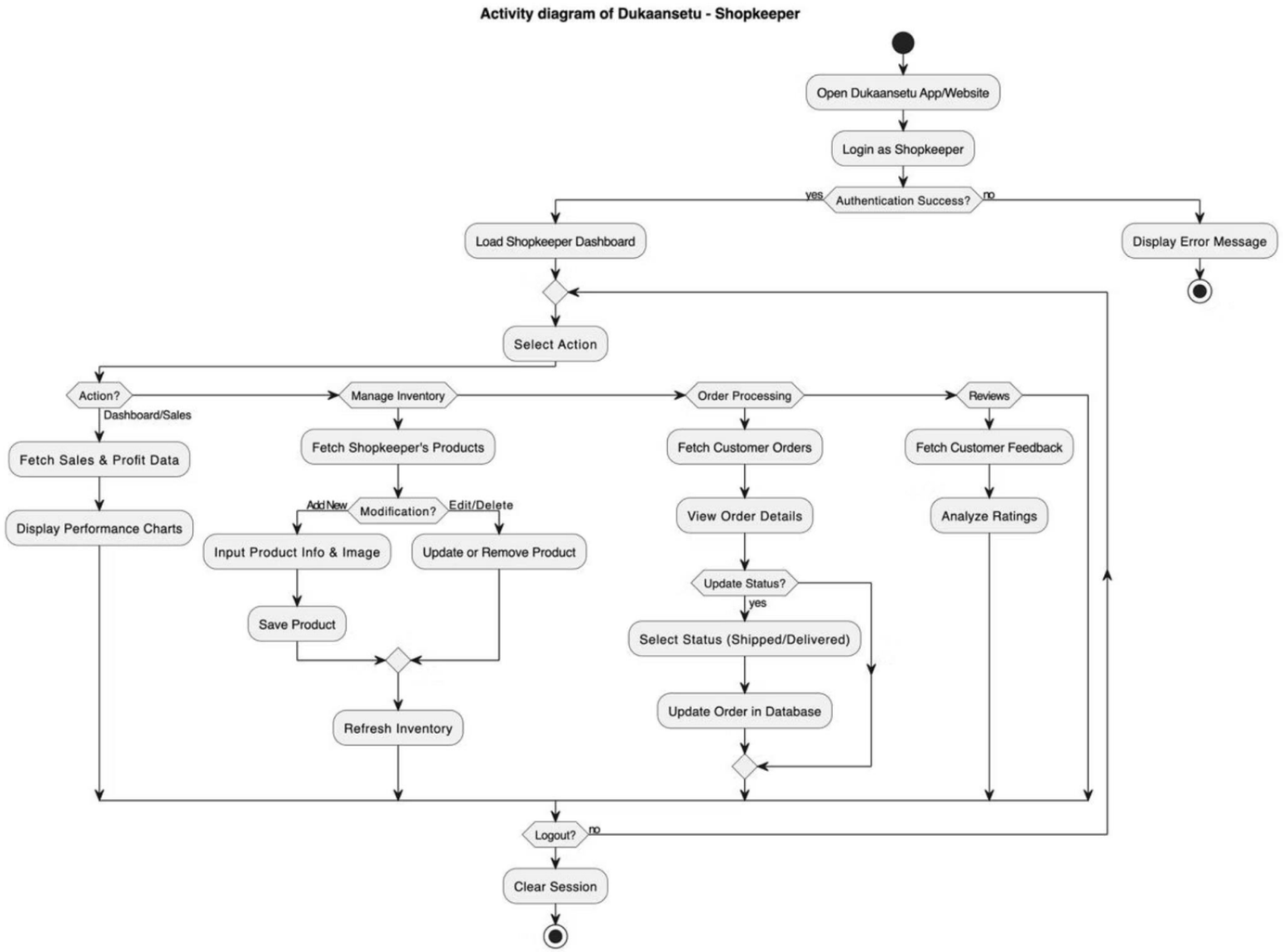

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Activity Diagram

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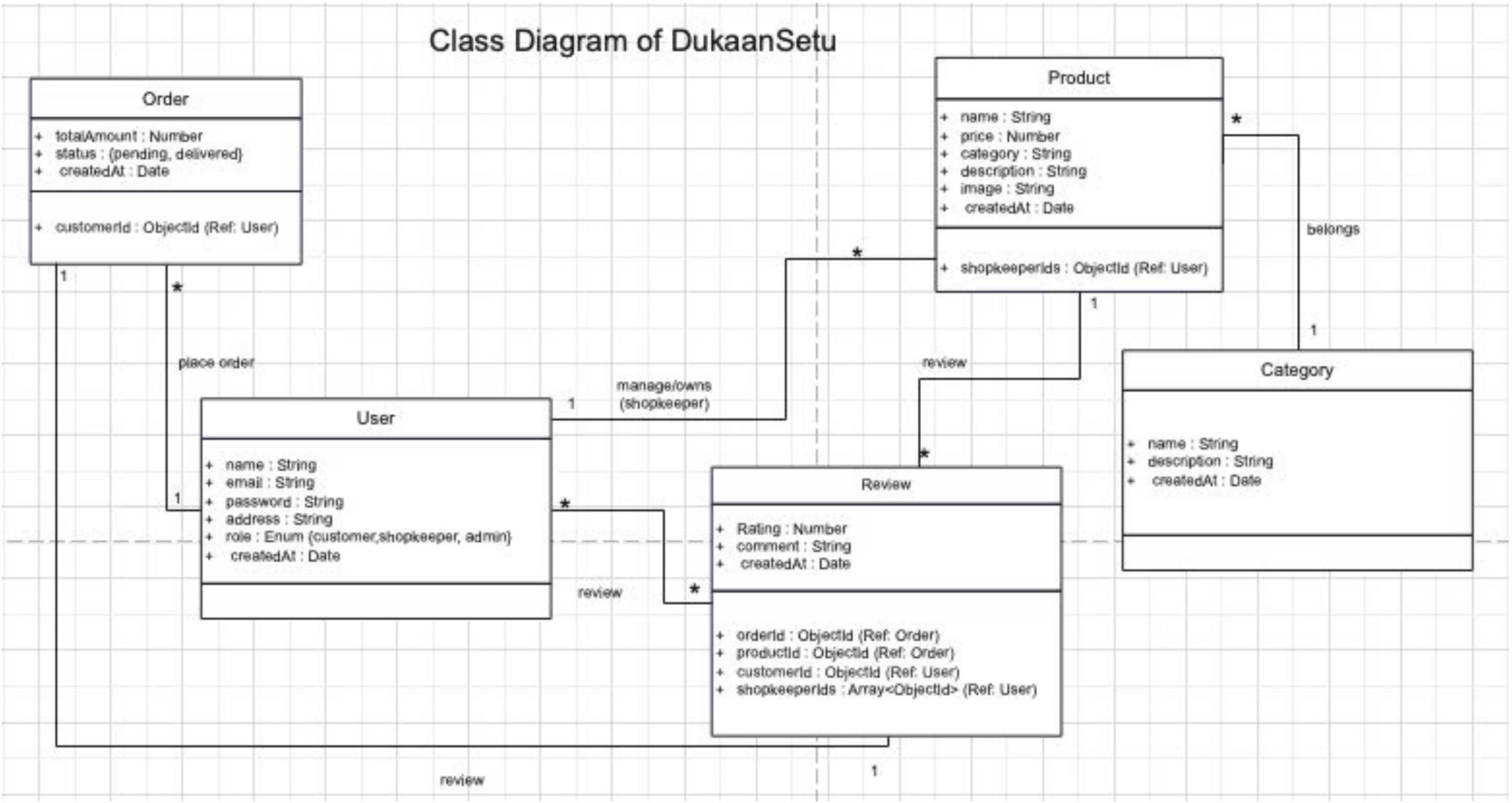
Class Diagram



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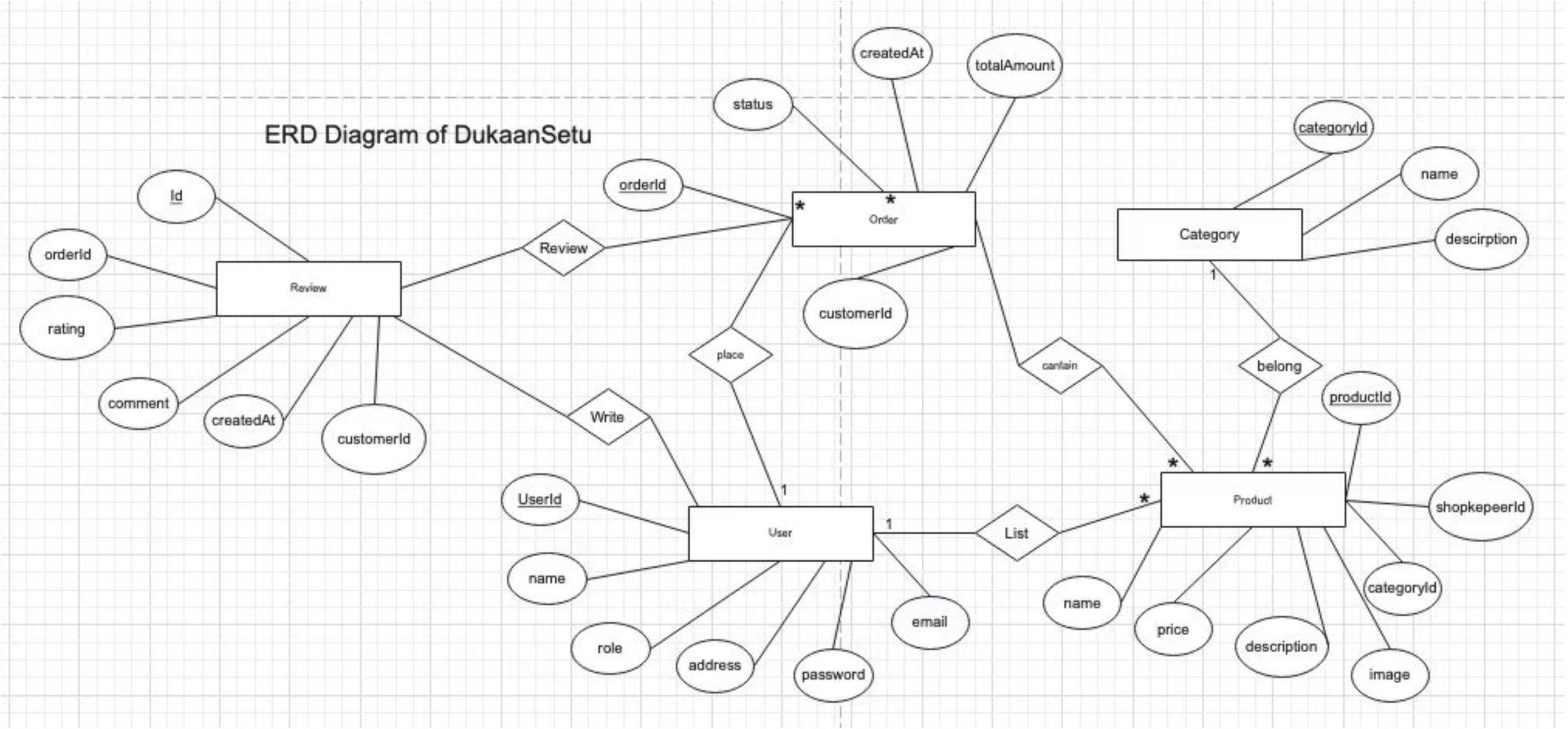
ERD Diagram



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Thank You!

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